

Ravi Tomer

Software Engineer — AI Platforms & Backend Systems

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Summary

Software Engineer with experience building AI platforms, distributed backend systems, and production-grade GenAI applications. Skilled in LLM orchestration, RAG architectures, GPU-optimized inference pipelines, and scalable API development using Python, FastAPI, LangGraph, OpenAI, and AWS. Focused on designing reliable, extensible systems that bridge AI capabilities with production software engineering.

Skills

Backend & Distributed Systems

- Python, FastAPI, Flask, Node.js, Express.js, REST APIs, GraphQL

Generative AI & LLM Engineering

- OpenAI APIs, GPT Models, LangChain, LangGraph, AI Agents, Ollama (Local LLM)
- RAG Architectures, Prompt Engineering, Structured Outputs, Vector Embeddings
- Transformers, Hugging Face, Vector Search, Agentic Workflows
- Prompt Optimization, Synthetic Data, Weaviate, vLLM, Llama

Data Platforms & Storage

- PostgreSQL, MongoDB, Vector Databases, Embedding Pipelines, FAISS

Cloud, Infrastructure & MLOps

- AWS, Firebase Functions, Docker, CI/CD Pipelines, Linux
- Model Deployment, Inference Optimization, GPU-Aware Workloads, ONNX

Architecture & Engineering

- Distributed Systems, Microservices, System Design
- Async Programming, Caching, Design Patterns

Experience

Software Development Engineer

Aug 2025 – Present

Saralweb, Noida, India

- Built GPU-optimized AI inference pipelines, reducing image generation latency by **3×** while improving throughput and resource utilization.
- Architected a **multi-stage gated inference framework** that reduced false negatives by **~90%** without increasing infrastructure costs or inference load.
- Developed a deep learning-based face anti-spoofing solution using **MiniFaceNetV2**, **ONNX Runtime**, and multi-crop liveness validation to detect replay and printed-photo attacks across desktop and embedded environments.
- Collaborated with AI, backend, and product teams to develop production-grade GenAI platforms, contributing to architecture, deployment, and operational reliability.
- Built and integrated **LLM-powered workflows** using prompt orchestration, context management, and structured output pipelines to automate business processes.
- Engineered scalable backend services and distributed systems using Python and FastAPI, improving deployment efficiency, reliability, and maintainability.
- Leveraged AI-native development tools including OpenAI Codex and Claude Code to accelerate feature development, code generation, refactoring, debugging, and architecture iteration.

Software Developer

Jun 2024 – Jul 2025

Indevz, Noida, India

- Developed backend services and APIs for inventory management, billing, order processing, and return workflows.
- Built serverless event-driven applications using **Firebase Functions**, enabling real-time tracking and notification systems.
- Implemented secure authentication and authorization using **Firebase Authentication** with role-based access control.
- Designed and optimized database schemas, business logic, and API integrations to improve system performance and maintainability.
- Developed an AI-powered video generation and processing engine supporting multiple video providers through a unified, model-agnostic architecture, enabling seamless provider switching, scalable inference workflows, and rapid integration of new AI video models.

Projects

Modular AI Agent Platform

- Architected a provider-agnostic AI agent platform supporting OpenAI and Anthropic models through a unified abstraction layer, enabling dynamic model routing, failover, and seamless provider switching.
- Designed and implemented a multi-agent orchestration framework for building and executing AI workflows, with support for agent collaboration, tool execution, and stateful task management.
- Built isolated memory and context management systems for individual applications, ensuring persistent conversational state while preventing cross-application context leakage.
- Developed asynchronous inference pipelines for prompt orchestration, tool invocation, and response generation, improving execution efficiency and reducing end-to-end latency.
- Created a modular plugin architecture enabling rapid integration of new agents, tools, workflows, and LLM providers with minimal code changes.
- Implemented structured output generation, workflow routing, and execution tracing to improve reliability, observability, and maintainability of AI-driven applications.
- **Tech Stack:** Python, LangGraph, FastAPI, AsyncIO, OpenAI API, Anthropic Claude API

Education

Master of Computer Applications (MCA)	2025 – 2027
Indira Gandhi National Open University (IGNOU)	
B.Sc. Physical Science and Computer Science	2021 – 2024
University of Delhi	